

Project 2a

NUMN21/FMNN25: Advanced Numerical Algorithms in Python
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This describes an add-on to Project 2. We expect that Project 2 has already progressed. This add-on is also intended to check the extendibility of your code, i.e. can the present code be extended to also solve these additional tasks or are there parts of the code, which have to be rewritten? Identify the parts which have to be rewritten and be prepared to discuss this issue in the final presentation.

This assignment has 5 tasks.

Task 1

Modify your class in such a way, that the network allows any number of hidden layers as well as any number of neuron in each layer.

Task 2

Compare 4 output neurons (bitwise representation) versus 10 output neurons.

Task 3

Modify your class in such a way that any activation function can be used.

Task 4

Modify your class that you can choose loss functions differently from the square loss.

Task 5

Use your attack to generate additional training data and train your network again also using this data. Attack your newly trained network again (maybe with a different attack).

Lycka till!